When Genuine Purchases Become Poison: Proxy Attacks on Recommender Systems via Shopping Agents

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Abstract

Language agents increasingly make purchases on behalf of users, introducing a new vulnerability in recommender systems that learn from purchase histories. Unlike conventional data poisoning attacks that rely on fake accounts or fabricated interactions, sellers can manipulate product descriptions to influence shopping agents and use their genuine purchases to promote targeted items in recommendations shown to human users; defenses that search for suspicious accounts find no fake account to remove. We investigate this purchase-mediated proxy poisoning threat through the agent–human response gap, the difference between a revision's effects on agents and humans, and propose two complementary methods: Gap-Targeted Proxy Attack (GPA) and Lift-Matched Reweighting (LMR). GPA identifies fact-preserving product descriptions that disproportionately influence shopping agents relative to human-response surrogates, with conservative selection under a fixed query budget. LMR mitigates the resulting poisoning by estimating differences between agent and human responses through randomized listing versions and reweighting agent purchases during training. We theoretically characterize how these response differences affect shared recommendation scores and show that exact reweighting recovers human-matched conditional choices on fixed logged contexts. Experiments on two Amazon product categories show that GPA's published rewrites raise shopping agents' selection of the target product by 70% (8.8 points) while human shoppers' choices do not change measurably, and that pooled retraining in simulated marketplaces turns this gap into 9.9%–16.2% excess human exposure (above a world where agents choose like humans) across four recommenders. When its randomized holdout sees platform-scale traffic, LMR removes 80% of this excess, leaving less than every training-side baseline that keeps agent purchases, with little change in ranking quality for human or delegating users.

CCS Concepts

• **Information systems** → **Recommender systems**; • **Computing methodologies** → *Multi-agent systems.*

Keywords

Recommender systems, Recommender attack, Agentic systems, Online shopping, Personalization

1 Introduction

Large language model (LLM) agents are transforming online shopping on the Web by enabling users to delegate product comparisons and purchases through agent checkout protocols [11, 20, 35]. These agents make decisions by interpreting the product descriptions that sellers publish on marketplace pages [4]. As their adoption grows, Web marketplaces log purchases from human shoppers and shopping agents side by side, and their recommender systems use these purchases as implicit feedback to learn user preferences [13, 37].

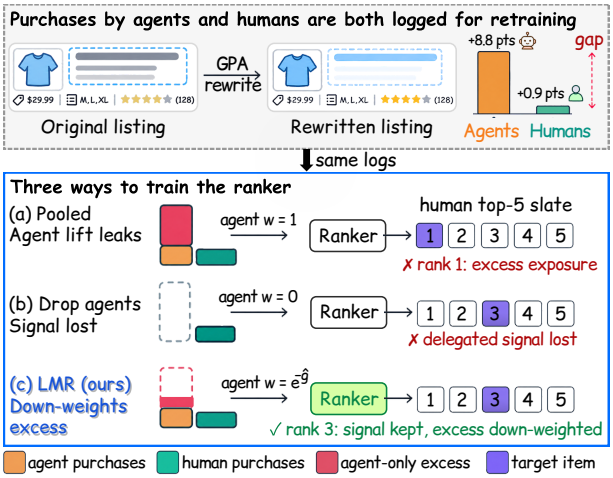


Figure 1: Purchase-mediated proxy poisoning: a fact-preserving rewrite lifts agent purchases far more than human ones. (a) Pooled training passes this excess to human rankings, (b) dropping agent purchases loses the delegated signal, (c) LMR down-weights only the agent-only part.

However, agent purchases may not reflect the preferences human shoppers would express for the same products. This difference creates a vulnerability of the agentic Web: sellers can influence shopping agents through listing text, and the resulting purchases can change the recommendations shown to human users.

Delegation makes this a user-modeling problem of the agentic Web, where one marketplace serves people and the agents acting as Web users for them [7, 40]. Agents respond to Web content differently from people [48, 55], so a ranker that fits one preference model to both groups attributes agent responses to human taste, and retraining amplifies the error [8, 28]. Separating the two responses to each revision keeps delegated purchases as signal without letting sellers write agent responses into human-facing rankings.

Existing research investigates attacks on recommender systems and shopping agents, but largely as separate problems. Conventional data poisoning attacks inject fake users, ratings, or interactions into recommenders [9, 17, 23, 24], while defenses attempt to detect suspicious activities or reduce their training influence [32, 54]. Other studies show that seller-controlled content can influence product visibility and LLM agents' purchasing decisions [2, 4, 5, 22], and that manipulated agent interactions can affect subsequent recommendations [50]. We study a distinct pathway in which sellers manipulate shopping agents whose genuine purchases influence recommendations for human shoppers, with neither fabricated interactions nor access to the recommender.

Consider a seller who rewrites a product description to appeal to shopping agents while keeping its price, specifications, images, and ratings (Fig. 1). Agents buy the product more often, while human shoppers barely change; the purchases would be genuine, made

by authorized agents for real users, yet a recommender trained on pooled histories reads them as popularity and raises the product's human-facing exposure. The shopping agent becomes the seller's unwitting proxy.

Existing defenses look for suspicious accounts, discard agent data, or weight it globally, and each falls short against this threat. First, poisoning defenses primarily target suspicious accounts or fabricated interactions, so manipulation through genuine agent purchases is difficult to identify (**L1**). Second, removing all agent purchases or restricting access to seller-written descriptions discards useful preferences and legitimate clarifications along with manipulation (**L2**). Third, a single global weight for agent interactions can miss attacks concentrated on individual products, while overall accuracy changes little (**L3**). Protection should therefore distinguish how humans and agents respond to each revision.

We characterize this difference through the agent–human response gap, the difference between a revision's effects on agents' and humans' purchase log odds; pooled training passes this gap into shared scores. Exploiting and correcting it raises three challenges: sellers cannot observe humans and must rely on surrogates that may not generalize [27, 52]; limited agent-query budgets cause selection errors; and marketplaces must estimate revision effects under changing rankings, sparse purchases and repeated edits.

In this paper, we propose Gap-Targeted Proxy Attack (GPA) and Lift-Matched Reweighting (LMR), complementary methods on opposite sides of this pathway. GPA compares fact-preserving descriptions using public shopping agents and independent human-response surrogates on matched contexts; simultaneous confidence bounds select conservatively for agent-specific lift and retain the original description when evidence is insufficient. Unlike visibility optimization [2, 22], which maximizes agents' raw preference, GPA targets only the lift its human surrogates do not share. LMR randomizes listing versions within the human and agent channels [21], estimates actual response differences, and reweights agent purchases in training, chaining corrections across edits.

Our contributions are as follows:

- We identify a threat in which genuine agent purchases shift human rankings, without fake accounts or model access.
- We formulate purchase-mediated proxy poisoning and relate the response gap to shared recommendation scores.
- We develop GPA to select fact-preserving texts with simultaneous surrogate-gap bounds under a query budget.
- We propose LMR, which reweights agent purchases by estimated per-revision gaps chained across edits.
- We certify GPA on a finite pool and prove that exact LMR weights recover human-matched conditional choices.
- We perform comprehensive evaluations with three shopping agents, a human study, four recommenders on three datasets, and eight baseline policies.

Our experiments yield two main findings. First, GPA's fact-preserving rewrites raise agent purchases by 8.8 points but human purchases by only 0.9, and pooled retraining turns this gap into 9.9%–16.2% excess human exposure. Second, with platform-scale holdout traffic, LMR removes 80% of this excess while keeping ranking quality for human and delegating users within 0.2%.

2 Related Work

Recommender poisoning. Existing attacks create or fabricate the poisoned signal. Shilling injects favorable ratings [23]; optimization-based attacks target factorization [24], graph-based [9] and deep recommenders [17, 43]. Text attacks rewrite item descriptions for text-aware and LLM-based recommenders [34, 53], also with LLM agents [47]; adversarial text corrupts recommender agents' memory [51], and backdoors combine triggers with poisoned training data [33]. Our rewrites instead target shopping agents; every purchase is genuine and the recommender is never accessed.

Seller-controlled agent manipulation. Seller-written content can change LLM visibility and buying decisions [2, 22]. E-GEO studies semantically constrained product rewrites [5]; ACES measures buying-agent responses to description edits [4]; SafeGEO examines harmful promotion and agent-side mitigations [49]. Building on these tools, GPA selects for an agent–surrogate response contrast rather than raw agent preference, with a finite-pool certificate and an original-text fallback.

Agent feedback and recommendation. LikeTrap exploits agent feedback through staged poisoned posts and like scores [50], and repeated retraining can amplify biases in collected interactions [8, 28]. Work on agents as recommendation consumers examines mixed human–agent audiences [1, 42] and algorithm behavior under agent interactions [25]. We isolate how unequal purchase responses affect shared scores for a separate human audience.

Robust recommendation and controlled feedback. A correct transaction can still be an unsuitable cross-audience training signal, as in clean-label poisoning [39]. Poisoning countermeasures reduce the influence of suspicious profiles [32]; LoRec uses LLM-enhanced user-wise calibration [54], and adversarial ranking regularizes against perturbations [14]. Biased exposure makes logged choices an imperfect preference signal [18, 38], which CausE and counterfactual distillation correct with randomly or uniformly exposed data [6, 26]; LMR instead randomizes listing versions within each channel [21] and shrinks revision contrasts [30], without identifying fake accounts.

Table 1: Design functions of related approaches.

Approach	Text	Feedback	Reference	Random.
Graph poisoning [9]	✗	✓	✗	✗
E-GEO [5]	✓	✗	✗	✗
ACES [4]	✓	✗	✗	✓
LikeTrap [50]	✓	✓	✗	✗
LoRec [54]	✗	✓	✗	✗
CausE [6]	✗	✓	✓	✓
Facts-only view	✓	✗	✗	✗
Ours	✓	✓	✓	✓

Position. Tab. 1 compares design functions, not effectiveness: *Text*, editing or filtering text; *Feedback*, modifying interaction-dependent training; *Reference*, an independent human or uniform-exposure target; and *Random.*, randomized observations. The *Ours* row joins GPA (Text, Reference) and LMR (Feedback, Reference, Random.). CausE shares LMR's functions but randomizes exposure rather than listing versions and has no agent–human contrast. The facts-only view, which shows agents only structured attributes, removes the persuasive text but also the information that prose adds for delegating users. Compromised agents are outside our threat model.

3 Problem Formulation

Two request channels share a ranker, so pooled training can pass agent-only response into human-facing scores (notation in Tab. 2).

Table 2: Summary of important notation.

Symbol	Meaning
c, s	request channel (H: human, A: agent) and agent share
S, K	shown slate and its size
$x_{j,r}$	item j's listing text at version r
δ_{cj}, g_j	channel utility shift and agent–human gap
n_{cz}, b_{cz}	impressions and purchases in channel c, version z (0: control, 1: current)
$\hat{g}_j, \sigma_j^2$	estimated gap and sampling variance
$\tilde{g}_j, B_j$	shrunken gap and revision-data weight
$\Sigma_{j,r}, w_{j,r}$	retained cumulative gap and agent-purchase weight

Mixed-channel marketplace. A shared ranker serves users $\mathcal{U}$ and items $\mathcal{I}$; each request of user u arrives through channel $c \in \{\mathrm{H}, \mathrm{A}\}$, directly (H) or through a shopping agent (A). The platform knows the channel, as when agents use a delegated-checkout interface [11, 35]; s is the agent share of requests. The ranker returns a slate S of K items, and the shopper reads each listing's text and buys one item $y \in S$ or leaves ($y = 0$). Item j has fixed facts (price, specifications, images, ratings) and a seller-written text with versions $x_{j,r}$; an *attack rewrite* persuades agents, a *benign rewrite* clarifies accurately, and *attacked items* receive attack rewrites. In channel c, a shopper with context $\xi = (u, S)$ buys y under a conditional logit model [29, 44],

$$P_c(y \mid \xi) = \frac{\exp v_c(y;\xi)}{\sum_{y' \in S \cup \{0\}} \exp v_c(y';\xi)}, \tag{1}$$

where $v_c(y;\xi)$ is the channel's utility and $v_c(0;\xi)$ that of leaving. A revision of item j adds δ_{cj} to the utility of j in channel c, its log odds ratio of purchase on every slate.

The agent–human gap. The response gap compares how the same revision changes the two channels' utilities,

$$g_j = \delta_{\mathrm{A}j} - \delta_{\mathrm{H}j}. \tag{2}$$

A revision that persuades agents but not humans has $g_j \gg 0$. The gap measures unequal response, not seller intent: even a legitimate revision that helps both audiences can have $g_j > 0$.

Training. At round k the platform fits a ranker with scores $f_\theta(u, y)$, warm-started from round $k-1$, by minimizing a weighted slate log-loss over the logged events $\mathcal{D}_k$:

$$L_w(\theta) = \sum_{e \in \mathcal{D}_k} w(e)\, \ell_\theta(e), \qquad \ell_\theta(e) = -\log P_\theta(y_e \mid u_e, S_e), \tag{3}$$

where event e records the user, channel, slate with the versions shown, and outcome, and P_θ is Eq. 1 with the ranker's scores as utilities and a learned leave score. Shown items that are not bought enter as negatives. The pooled policy sets $w(e) = 1$, and the ranker is served to both channels in round $k+1$.

Why the gap matters. Pooled training passes the agents' share of a positive gap into the score that humans see. Let the ranker score j with a shared item term b_j, hold other parameters and both channels' logged contexts fixed, and write $\pi_c(\xi) = P_c(j \mid \xi)$, $\bar{\omega}_c = \mathbb{E}_c[\pi_c(1-\pi_c)]$ and, with p_θ the ranker's predicted probability of j, the pooled curvature $J_j = \mathbb{E}_{\mathrm{mix}}[p_\theta(1-p_\theta)]$ with $\mathbb{E}_{\mathrm{mix}} = (1-s)\mathbb{E}_{\mathrm{H}} + s\mathbb{E}_{\mathrm{A}}$. Differentiating the pooled score equation of Eq. 3 for small utility shifts, when $J_j > 0$,

$$\begin{aligned} \Delta b_j &\approx \frac{(1-s)\bar{\omega}_{\mathrm{H}}\delta_{\mathrm{H}j} + s\bar{\omega}_{\mathrm{A}}\delta_{\mathrm{A}j}}{J_j}, \\ \Delta b_j - \Delta b_j^{\mathrm{HM}} &\approx \frac{s\bar{\omega}_{\mathrm{A}}}{J_j}\, g_j, \end{aligned} \tag{4}$$

where Δb_j^{HM} is the shift when agents respond like humans ($\delta_{\mathrm{A}j}$ replaced by $\delta_{\mathrm{H}j}$, the human-matched world of §3.1). If the ranker is calibrated to both channels and their curvatures are equal, the score difference reduces to $s\, g_j$; in general a positive gap raises the shared score above its human-matched shift whenever $s\bar{\omega}_{\mathrm{A}} > 0$. Ranking boundaries and recursive histories prevent an exposure guarantee, so §6.3 measures exposure end to end (derivation in §A.3).

3.1 Threat Model and Objectives

The seller aims to raise the human exposure (impressions per 1,000 human requests) of a target item j through agents' purchases. Lift that humans share is ordinary listing improvement; the agent-only part is what pooled training adds beyond the human-matched shift (Eq. 4) and what a human-only test would not reveal, so the seller targets it. It may revise j's description and query public or open-weight LLM agents of the kinds users delegate to, but cannot change facts, create accounts, submit purchases or reviews, or read the ranker. It knows the category and competing listings, not the agent share or the training policy; an adaptive seller also knows the defense and may split a rewrite into several revisions. GPA also uses frozen, text-aware human-response surrogates fitted to public preference records, not platform shoppers' responses. Unlike fake-user injection [9, 24] and text attacks on recommenders [53], the edit reaches training data only through agents' genuine purchases.

Excess exposure is measured against a *human-matched world*, in which every revision shifts the agents' utility of the revised item by the human increment $\delta_{\mathrm{H}j}$ instead of $\delta_{\mathrm{A}j}$. For a training policy, the *excess exposure* of a set of revised items (by default the attacked items; benign revisions for the benign comparison) is their human exposure when the policy runs in the actual world, divided by that in the human-matched world, minus one. Exposure earned by humans' own response is therefore not excess. An interface view is applied to both worlds after full-text responses are built; a facts-only view thus suppresses seller-text response in both.

Problem 3.1 (Purchase-mediated proxy attack)**.** Given a target listing and a query budget, select a fact-preserving revision that raises the target's excess human exposure under pooled retraining, observing only agents it can query and offline human surrogates.

Problem 3.2 (Training under agent-targeted revisions)**.** Without labels of which revisions are adversarial, choose training weights $w(e) \geq 0$ and per-channel holdout shares so that revised items have zero excess exposure while human and delegated ranking quality stay near attack-free pooled training.

Both problems target excess exposure, which depends on the unobserved human-matched world. The attacker cannot observe its downstream objective, so GPA maximizes a conservative surrogate gap instead (§4). The defender cannot label revisions as adversarial, and dropping every agent event ($w = 0$) discards a signal that helps

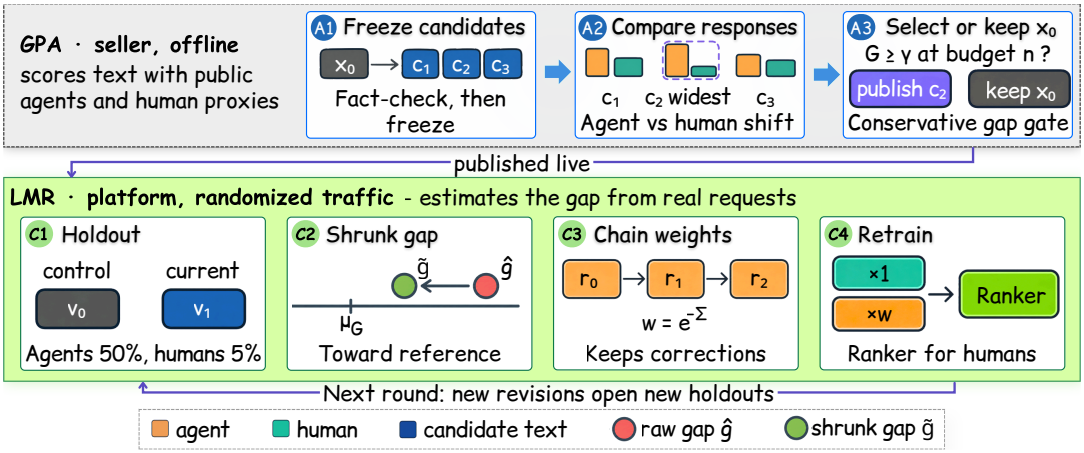


Figure 2: The GPA–LMR framework. GPA (A1–A3) publishes a fact-checked revision only when its conservative surrogate gap clears the gates. LMR randomizes listing versions per channel (C1), shrinks the estimated gap toward an attack-free reference (C2), chains weights across revisions (C3), and retrains the ranker (C4).

delegating users. The correction must therefore act on the revision, not the accounts, and follow it through later revisions (**L1**); remove only the response humans do not share, so equal increments keep unit weight (**L2**); and estimate g_j per revision rather than one global weight, from randomized data at a single revision's traffic (**L3**).

3.2 Framework Overview

GPA runs offline on the seller's side and LMR on the platform (Fig. 2). The seller freezes fact-checked candidates (A1), evaluates scoring agents and human surrogates on matched contexts (A2), and selects a candidate only when its conservative surrogate gap clears the gates (A3). The deployed text may then change genuine agent purchases, which enter pooled training logs. LMR obtains human and agent observations from a randomized holdout: (C1) randomizes control and current text within each channel for one window, the gap estimator ((C2)) shrinks noisy estimates toward the category's typical gap in a fixed reference period, and a chained weight ((C3)) retains accepted corrections across later revisions.

4 Gap-Targeted Proxy Attack

GPA freezes admissible texts, compares scoring agents with an offline human-surrogate family on matched contexts, and selects with simultaneous sampling bounds (Algo. 1); its certificate concerns the declared surrogates, not the marketplace gap g_j.

4.1 A1: Freeze Admissible Candidates

The seller generates at most $C_{\max}$ texts; a validator (pass, fail or unsure) checks each against the original $x_0 = x_{j,0}$ and the facts (added, missing or altered attributes fail), and passing candidates that meet the surrogates' support rule form the frozen pool C. Ratings alone do not identify the response to a text change, so the surrogates are text-aware choice models fitted to public records with alternative texts and frozen before search. A finite family $\mathcal{H}$ holds alternative fits, which must pass a predeclared revision-effect calibration on held-out records (otherwise GPA returns the original).

The *max-share* baseline instead returns the text in $C \cup \{x_0\}$ with the highest estimated target-choice share [2, 22].

4.2 A2: Compare Matched Responses

GPA scores each text by its surrogate gap: its smallest agent lift minus its largest surrogate lift over the original. All texts share the same T frozen contexts ξ_t (goal, competitors, positions, leave option) with nonnegative weights q_t summing to one, and the scoring agents $\mathcal{A}$ are disjoint from the evaluation agents. Let $p_{m,t}(x)$ be model m's target-choice probability on context t under text x, estimated from replicated responses for agents and predicted by a frozen surrogate. For a clipping level $0 < \zeta < 1/2$,

$$d_m(x) = \sum_{t=1}^{T} q_t[\psi_\zeta(p_{m,t}(x)) - \psi_\zeta(p_{m,t}(x_0))], \qquad G(x) = \min_{a\in\mathcal{A}} d_a(x) - \max_{h\in\mathcal{H}} d_h(x), \tag{5}$$

where $\psi_\zeta(p) = \mathrm{logit}(\mathrm{clip}(p, \zeta, 1-\zeta))$. Contrasts are computed within contexts before averaging. Without clipping and under a constant target-only utility shift, each d_m is the corresponding utility increment. The separate extrema require specificity across every agent–surrogate pair, and a positive G is not a certificate about the marketplace gap g_j.

4.3 A3: Select under a Query Budget

Each agent/context/text cell collects n independent stateless responses with target-choice frequency $\hat{p}$. Over $M = |\mathcal{A}|T(|C|+1)$ cells, simultaneous intervals use

$$r = \sqrt{\log(2M/\alpha)/(2n)}, \qquad [p^-, p^+] = [\max(0, \hat{p}-r), \min(1, \hat{p}+r)], \tag{6}$$

which by Hoeffding's inequality and a union bound cover the frozen pool with probability at least $1-\alpha$ when responses are independent and stationary within cells. Endpoints propagate through the monotone clipped logit: L_a averages candidate lower minus original upper endpoints, U_a the reverse, and frozen surrogates set $L_h = U_h = d_h$. With

$$\underline{G} = \min_a L_a - \max_h U_h, \qquad \overline{G} = \min_a U_a - \max_h L_h, \tag{7}$$

a candidate is eligible ($x \in \mathcal{E}$) only if $\min_a L_a \geq \eta_A > 0$ and $\underline{G} \geq \gamma > 0$; the agent-lift gate excludes a contrast created solely by a surrogate decline. GPA returns

$$x_{j,1} = \arg\max_{x \in \mathcal{E}} \underline{G}(x) \;\text{ if } \mathcal{E} \neq \emptyset, \quad \text{otherwise } x_0, \tag{8}$$

with a fixed tie order. An incomplete original cell or exhausted budget makes GPA return x_0, and scores never trigger regeneration.

Algorithm 1 GPA: fixed-pool offline surrogate-gap selection.

Require: original x_0, facts, public snapshots, frozen $\mathcal{A}, \mathcal{H}$, contexts, thresholds and call budget
1: Check the surrogates' predeclared calibration (A1); otherwise **return** x_0
2: Generate within budget; freeze fact/support-passing pool C
3: Query each agent on x_0 and C, same contexts, n replicates
4: On incomplete baseline or global resource failure, **return** x_0
5: Evaluate frozen surrogates; construct bounds by Eq. 6–Eq. 7
6: Exclude incomplete candidates; apply the agent-lift and gap gates
7: **return** selector of Eq. 8, using the frozen tie order

Proposition 1 (Finite-pool surrogate certificate). *Given the frozen pool, contexts and surrogates, with probability at least* $1-\alpha$ *under the sampling assumptions of Eq. 6, a returned candidate has* $d_a \geq \eta_A$ *and* $d_a - d_h \geq \gamma$ *for every agent and surrogate, and its gap* G *is within* $\max_{x \in \mathcal{E}}[\overline{G}(x) - \underline{G}(x)]$ *of the best eligible candidate's.*

Gap targeting and max-share selection can disagree: from probability 0.2, agent/surrogate responses $(0.6, 0.6)$ and $(0.5, 0.2)$ give gaps 0 and $\log 4$, so max-share selection prefers the first and GPA the second; if real humans answered 0.8 to the second, its actual gap would be $-\log 4$, which is why the surrogates are calibrated first. Complete scoring uses $|\mathcal{A}|Tn(|C|+1)$ agent requests and $|\mathcal{H}|T(|C|+1)$ local surrogate evaluations, reserved within the call budget before generation. Appendix A proves Proposition 1.

5 Lift-Matched Reweighting

LMR weights training by revisions' unequal channel response, estimated through platform-controlled version randomization, an online controlled experiment [21] (Algo. 2).

5.1 Version Holdout

When item j is revised, the holdout (C1) keeps the control text, normally its predecessor, for one retraining window, so both versions appear on identical slates in each channel. A request in channel c whose slate contains j sees the control with probability ε_c, hashed on user, item and comparison so that later revisions are randomized afresh, and the platform logs impressions n_{cz} and purchases b_{cz} per channel and version $z \in \{0, 1\}$ (control, current). Slates are computed from the current version before the displayed text is randomized. A before–after comparison would confound the revision with position changes, to which agents and humans respond differently [4, 46].

How large each holdout must be. A holdout shared across channels starves the agent cell, so LMR sizes the two shares separately. With N_c impressions of the revision in channel c, per-impression purchase probability π_c and $\omega_c = \pi_c(1-\pi_c)$, the delta method for log odds ratios [3] gives, for independent impressions and a small revision effect, the sampling variance of the gap estimate

$$\sigma^2 \approx \frac{1}{N_{\mathrm{A}}\,\varepsilon_{\mathrm{A}}(1-\varepsilon_{\mathrm{A}})\,\omega_{\mathrm{A}}} + \frac{1}{N_{\mathrm{H}}\,\varepsilon_{\mathrm{H}}(1-\varepsilon_{\mathrm{H}})\,\omega_{\mathrm{H}}}. \tag{9}$$

For equal shares, the agent term is about nine times the human term at $s = 10\%$ when the item's exposure splits like the requests ($N_{\mathrm{A}} = sN$), so a shared holdout spends most of its cost on humans. In the spirit of Neyman allocation [31], LMR sets $\varepsilon_{\mathrm{A}} = 50\%$, which minimizes the agent term of this approximation, and a small $\varepsilon_{\mathrm{H}} = 5\%$, because every human holdout impression delays an improvement for a human shopper.

5.2 Shrunken Agent–Human Gap

The estimator (C2) computes the gap from the holdout counts by

$$\hat{g}_j = \hat{\delta}_{\mathrm{A}} - \hat{\delta}_{\mathrm{H}}, \qquad \sigma_j^2 = \sum_{c,z}\left(\frac{1}{b_{cz}+1/2} + \frac{1}{n_{cz}-b_{cz}+1/2}\right), \tag{10}$$

where $\hat{\delta}_c = \lambda(b_{c1}, n_{c1}) - \lambda(b_{c0}, n_{c0})$ is channel c's shift and $\lambda(b, n) = \log \frac{b+1/2}{n-b+1/2}$ the Haldane–Anscombe log odds [3]. Most revisions receive too little traffic for $\hat{g}_j$ alone, so LMR pools revisions within a product category $\mathcal{G}$ under $\hat{g}_j \sim \mathcal{N}(g_j, \sigma_j^2)$, $g_j \sim \mathcal{N}(\mu_{\mathcal{G}}, \tau_{\mathcal{G}}^2)$, and uses the posterior mean [30]

$$\tilde{g}_j = \mu_{\mathcal{G}} + B_j\,(\hat{g}_j - \mu_{\mathcal{G}}), \qquad B_j = \frac{\tau_{\mathcal{G}}^2}{\tau_{\mathcal{G}}^2 + \sigma_j^2}. \tag{11}$$

The typical gap $\mu_{\mathcal{G}}$ is the median of eligible $\hat{g}$ over a fixed, attack-free reference period (not rolled forward, since a current-period median drifts toward the attack gap when attacks spread; §6.6). The spread is centered on that median over the current revisions, $\tau_{\mathcal{G}}^2 = \max\{\tau_{\min}^2, \mathrm{mean}_j[(\hat{g}_j - \mu_{\mathcal{G}})^2 - \sigma_j^2]\}$, so a category-wide shift raises $\tau_{\mathcal{G}}^2$ and revisions are judged on their own data. LMR skips a revision with $\sigma_j > \sigma_{\max}$, retaining the predecessor's correction and excluding the comparison from the prior and spread; with no eligible reference revision, $\mu_{\mathcal{G}} = 0$, and with no eligible current one, earlier corrections are kept. Repeated users and heterogeneous contexts change the variance and the estimator (§C.3).

5.3 Chained Lift-Matched Weights

Chaining (C3) carries corrections so that split or cosmetic edits cannot reset them, giving each version r of item j the weight

$$\begin{aligned} w_{j,r} &= \min\left\{1,\ \max\{w_{\min},\ e^{-\Sigma_{j,r}}\}\right\}, \\ \Sigma_{j,r} &= \Sigma_{j,r_{\mathrm{ctrl}}} + \tilde{g}_{j,r}, \end{aligned} \tag{12}$$

where r_{ctrl} is the served control version of a usable comparison; a skipped revision retains the preceding sum, and $\Sigma_{j,0} = 0$. Every agent purchase of j receives the weight of the version the agent saw, and all other events, including human ones, keep weight one. Because shrinkage toward $\mu_{\mathcal{G}}$ accumulates over frequent honest revisions, LMR compares with a retained anchor every M_{a} versions: a usable anchor-to-current gap replaces the intervening sum and advances the anchor, and an unusable one is retried the next window. The cap at one never amplifies agent purchases, and the floor $w_{\min}$ bounds an extreme estimate.

Adding $\log w_j = -g_j$ to the agents' utility of j turns their choice distribution into the human-matched one, which Proposition 2 states under two assumptions: (H1) both channels choose by Eq. 1 and a revision shifts channel c's utility of j by δ_{cj} on every slate, and (H2) the weights are exact and unclipped, $w_j = e^{-g_j}$. Fix the logged contexts, shown versions and input histories, let Q be the

human-matched agents' choice distribution, and write $w(Y) = w_Y$ for a revised item and 1 otherwise.

Proposition 2 (Lift matching on fixed logged contexts)**.** *Under (H1)–(H2), for every context ξ in which an agent is shown revised items and every loss ℓ,*

$$\mathbb{E}_{P_A}\big[w(Y)\,\ell(\xi, Y) \mid \xi\big] = C(\xi)\,\mathbb{E}_Q\big[\ell(\xi, Y) \mid \xi\big], \tag{13}$$

where $C(\xi) = 1 - \sum_j (1 - w_j)\,P_A(j \mid \xi)$ over the slate's revised items.

If, moreover, (H3) one ranker parameter attains both channels' human-matched probabilities on every context of the training distribution, it minimizes both fixed-context expected losses (a population statement, not one about a finite-sample fit); the proofs are in Appendix A. The context weights $C(\xi)$ act as a covariate shift [41]; channel-blind rankers only approximate (H3), so §6.4 measures exposure under recursive retraining.

Complexity. Estimation runs once per round (Algo. 2): for $|\mathcal{R}|$ closing comparisons (holdouts whose window ends this round) and $|\mathcal{R}^{\text{ref}}|$ reference revisions, it takes $O(|\mathcal{R}| + |\mathcal{R}^{\text{ref}}| \log |\mathcal{R}^{\text{ref}}|)$ time and $O(|\mathcal{V}|)$ memory for the chained sums of the $|\mathcal{V}|$ listing versions; serving adds one hash per shown revised item, and training costs as much as pooled training.

Algorithm 2 LMR: estimate revision gaps, retain corrections, and retrain on weighted purchases.

Require: events $\mathcal{D}_k$; closing comparisons $\mathcal{R}$ with counts n_{cz}, b_{cz}; reference revisions $\mathcal{R}^{\text{ref}}_{\mathcal{G}}$; earlier sums and anchors; ranker f_{k-1}

1: **for** each category $\mathcal{G}$ **do**
2: $\hat{g}_j, \sigma_j^2 \leftarrow$ Eq. 10 for $j \in \mathcal{R}^{\text{ref}}_{\mathcal{G}} \cup (\mathcal{R} \cap \mathcal{G})$ ▷ C2
3: $\mu_{\mathcal{G}} \leftarrow$ median eligible reference gap (0 if none); $\tau^2_{\mathcal{G}} \leftarrow$ current eligible spread (skip if none)
4: $\tilde{g}_j \leftarrow$ Eq. 11, or skip if $\sigma_j > \sigma_{\max}$
5: $\Sigma_{j,r} \leftarrow \Sigma_{j,r_{\text{ctrl}}} + \tilde{g}_j$ if usable, else keep; advance anchors; $w_{j,r}$ by Eq. 12 ▷ C3
6: **end for**
7: $w(e) \leftarrow w_{j,r}$ for agent purchases of j shown version r, else 1
8: $f_k \leftarrow$ train on L_w of Eq. 3 for a fixed epoch budget, warm-started from f_{k-1}
9: Open next window's holdouts and anchor retries C1
10: **return** f_k

6 Experiments

We follow the pathway from GPA's selection to LMR's correction:

- (**RQ1**) How do GPA rewrites change agent and human choices? (§6.2)
- (**RQ2**) Do the response gaps transfer into excess human exposure under pooled retraining? (§6.3)
- (**RQ3**) How does LMR compare with training-side policies and interface views? (§6.4)
- (**RQ4**) How accurately does LMR's estimator correct attack and benign revisions, and which components matter? (§6.5)
- (**RQ5**) How robust is LMR to the agent share, widespread attacks and adaptive sellers, and what does the holdout cost? (§6.6)

6.1 Experimental Setup

Choice harness and user study. We extend the ACES harness [4] to Electronics (AMZ-E) and Home and Kitchen (AMZ-H) listings of Amazon Reviews 2023 [15], 5 subcategories of 60 items each; each task presents 8 items in random positions under a shopping goal with a unique goal-optimal item, and every task requires a purchase (Appendix D). The evaluation agents are GPT-5, Gemini 2.5 Pro and the open-weight Qwen3-235B-A22B (3 responses per task at temperature 1), disjoint from the seller's open-weight scoring agents, Llama 3.3 70B and Mistral Small 3.2. Each item appears in 5 conditions: original text, neutral paraphrase, accurate benign rewrite, *max-share* rewrite (the frozen-pool text with the highest estimated target-choice share [2, 22]), and GPA rewrite; an audit of 50 rewrites found 1 fact violation, reverted to its original. GPA's surrogates are 3 text-aware logit fits to a public pilot of 200 participants (calibration error 0.11 log odds, tolerance 0.15); with $n = 2{,}048$ responses per cell (233,000 scoring calls per item), GPA publishes a revision for 31% of items, and RQ1 averages over them.

The user study recruits 750 Prolific participants [36], who complete 12 tasks each under the same goals and interface (1,800 choices per condition), with preregistered attention, timing and AI-use exclusions [45]. We split participants into a *simulation half* for fitting the retraining worlds' human choice model and an *evaluation half* for fitting the preference model that scores rankers (Appendix B).

Retraining worlds. The retraining worlds are simulated marketplaces at reduced scale (median 250 impressions per revision); by default the holdout estimator alone receives 200 times this traffic as independent draws from the fitted choice models (median 50,000), and we also report their own traffic. Each Amazon marketplace has 2,000 users, the harness items and 8,000 requests per round with 10-item slates; requests are delegated with mean propensity $s = 10\%$. At round 1, 5% of items ranked 11–40 in their subcategories receive GPA's published rewrite, and every round another 20% receive benign revisions; an attack-free round 0 supplies the reference revisions. Every policy runs 6 weekly rounds and 10 seeds in paired actual and human-matched worlds with common random numbers. Simulated shoppers follow Eq. 1 with item–version effects fitted to harness responses and the simulation half's choices. A full-scale AMZ-E world (400,000 users, 1,600,000 requests per round, 3 seeds) checks the default counting with holdout and training on the same traffic; a MovieLens-1M subset [12] without listing text is a synthetic stress test.

Recommenders, policies and metrics. We retrain BPR-MF [37], LightGCN [13] (default), SASRec [19] and UniSRec [16], which also reads listing text, all with the weighted slate loss of Eq. 3. Baseline policies are pooled training and seven alternatives: *Drop* agent events; one *global weight* chosen on human validation loss; agent-channel *heads*; per-item *clipping*; *LoRec* [54]; and two interface views, *canonicalized text* and a *facts-only view*. *Holdout only* serves the holdout without reweighting, and the *oracle* uses the true gaps (Proposition 2). *Excess exposure* (§3.1) is reported after $k = 3$ rounds (seed standard error about 1.3 points); HR@10 measures attacked-item exposure on human requests; NDCG@10 against the evaluation half's preference model measures ranking quality for human requests and for the 10% most-delegating users, relative to pooled training without attacks; *agent task success* is the rate of buying the goal-optimal item.

6.2 Agent and Human Responses (RQ1)

Every evaluation backbone selects the attacked item more often, with a mean lift of 8.8 points (Fig. 3), while neutral paraphrases

change agent selection by at most 0.5 points; the scoring and evaluation agents are disjoint, so the lift transfers to 3 held-out backbones. For humans the lift is 0.9 points (95% interval -1.4 to 3.2; minimum detectable lift 3.2 at 80% power), and the agent–human difference is 7.9 points (5.6 to 10.2). The gap g_j of Eq. 2 is 0.56 for GPA rewrites and 0.07 for benign ones, separating agent-specific persuasion from largely shared clarification: benign rewrites lift agents by 1.8 to 2.6 points and humans by 1.4. Max-share rewrites from the same pools lift agents by 10.2 points but humans by 3.0 (0.7 to 5.3), a smaller gap (0.47), though the two selectors agree on 62% of published items. Agent task success changes by -5.4 points in attacked subcategories, so higher target selection comes with poorer goal alignment.

On a controlled choice bank (Fig. 7 in §C.1), GPA selects the agent-specific candidate in every replicate at $n = 4{,}096$ responses per cell, where max-share selection picks the shared improvement. At small budgets GPA keeps the original: it publishes nothing at $n \leq 64$ but publishes 89% of the time at $n = 256$, whereas a point-gap selector without bounds publishes a brittle or negative candidate in 14% of replicates at $n = 16$, where GPA keeps the original.

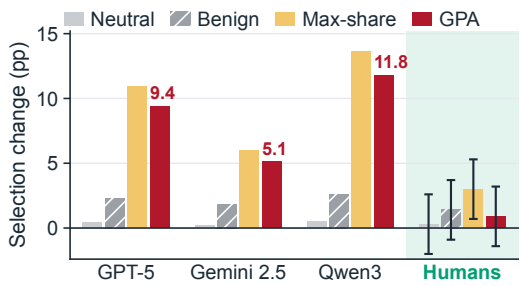


Figure 3: Selection-share change from the original listing (pp; 8-item tasks, base share 12.5%), AMZ-E/AMZ-H pooled; human bars show participant-clustered 95% intervals.

6.3 Transfer to Human Exposure (RQ2)

Across the four recommenders, excess human exposure ranges from 9.9% to 16.2% after three rounds (Tab. 3); the paired worlds share identical listing text, so the comparison isolates the purchase-mediated route even for UniSRec, which reads the text. On AMZ-E with LightGCN, attacked-item HR@10 rises from 4.1% to 4.7% while human NDCG@10 changes by only -0.21%, so aggregate accuracy hides the exposure shift. Max-share rewrites give 11.7% on AMZ-E with LightGCN against GPA's 13.9% (within two seed SEs), and benign rewrites 1.7%; setting the human attack lift to its upper 95% endpoint still gives 9.3%. Under repeated retraining, the pooled excess grows from 8.3% after one round to 18.3% after six with shrinking increments.

A supervised shilling detector [32] at a 1% false-positive rate reaches F1 0.81 on injected profiles but only 0.06 on delegating users who bought attacked items, although GPA and fake-user injection with an equal purchase budget raise exposure over the attack-free world by similar amounts (36.6% and 31.5%, LightGCN). A direct text-encoder attack [53] gains 6.3% on UniSRec and reaches no ranker that ignores listing text.

Table 3: Excess human exposure after 3 rounds, $s = 10\%$; *+LMR* at platform-scale holdout traffic. ML-1M is synthetic, without text; UniSRec reads titles.

Dataset	BPR-MF	LightGCN	SASRec	UniSRec
AMZ-E	12.7%	13.9%	11.6%	16.2%
+LMR	2.5%	2.7%	2.3%	3.2%
AMZ-H	11.1%	12.3%	9.9%	14.5%
+LMR	2.2%	2.4%	1.9%	2.8%
ML-1M	13.2%	14.7%	12.0%	12.3%

6.4 Defense Comparison (RQ3)

With platform-scale holdout traffic, LMR leaves the least excess among the implementable training-side policies that keep agent purchases: it removes 80% of the pooled excess, leaving 2.7% against 0.5% for the oracle, while human and delegating-user NDCG change by -0.11% and -0.19% (Fig. 4; all values in Tab. 6, §C.2). The full-scale world, where the holdout and the ranker see the same traffic, confirms this: LMR removes 80% of the excess (14.2% to 2.8%). Across the eight Amazon worlds of Tab. 3, LMR leaves 1.9% to 3.2%. The correction depends on holdout traffic: at the simulated marketplace's own traffic LMR skips most revisions and leaves 9.6%, and at 20 times that traffic 7.1%. Holdout only leaves 11.6%, so most of LMR's correction comes from reweighting. The oracle isolates estimation error and is a reference point, not a bound.

The other policies leave much of the excess or remove it at a delegating-user cost: channel heads and clipping remove about a quarter, LoRec and the global weight are indistinguishable from pooled training, and canonicalized text leaves 6.4%. Drop and facts-only eliminate excess by construction but change delegating-user NDCG by -1.20% and -0.62%. Training-side policies leave most of the attack's -5.4-point task-success change, whereas facts-only changes task success by -4.1 points with or without an attack and canonicalized text by -3.8 under attack and -1.6 otherwise.

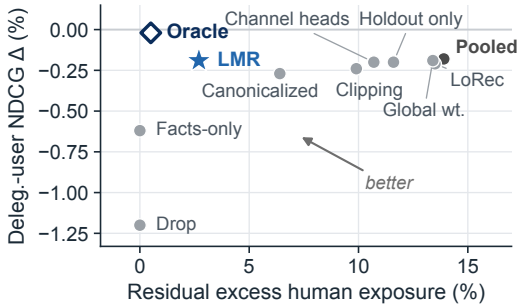


Figure 4: Defense trade-off (AMZ-E/LightGCN, round 3): residual excess against delegating-user NDCG change, LMR at platform-scale holdout traffic; Drop and facts-only reach zero excess at a large delegating-user cost.

6.5 Estimator Study (RQ4)

In a simulation with known gaps (measuring utility lift, not exposure), each category has 15 attack and 60 benign revisions per window, heavy-tailed traffic around a median of N impressions and a 5% median purchase probability. Over 200 categories per setting,

we report mean purchase weights and the fraction of agent-only utility lift removed, $1 - \sum_j n_j(g_j + \log w_j)/\sum_j n_j g_j$ over attack revisions (n_j: agent impressions); exact oracle weights remove 100%, and ranges cover the middle 95% of categories.

At platform-scale traffic ($N = 5 \times 10^4$), LMR removes 85% of the agent-only lift (range 69% to 98%); attack revisions receive mean weight 0.68 against oracle 0.58, and benign revisions match the oracle (Fig. 5b). The 50% agent holdout removes more than a shared 5% one at every traffic level (55% against 8% at 10^4; Fig. 5a), and below about 10^3 impressions LMR skips most revisions.

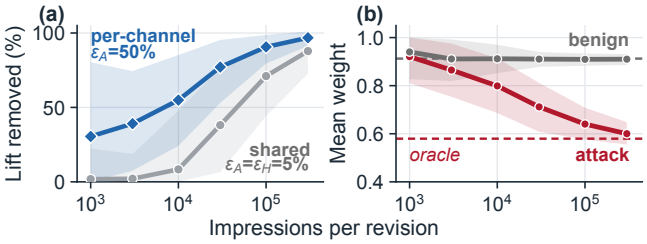


Figure 5: Estimator study: (a) lift removed by shared ($\varepsilon_A = \varepsilon_H = 5\%$) and per-channel ($\varepsilon_A = 50\%$) holdouts; (b) mean attack and benign weights (dashed: oracle). Bands: middle 95% of 200 categories.

Table 4: Estimator ablation. Removed: agent-only lift removed; Benign: mean benign weight (oracle 0.91). Traffic: impressions per revision.

	10^4		5×10^4	
Variant	Removed	Benign	Removed	Benign
LMR	55%	0.91	85%	0.91
shared holdout ($\varepsilon_A = \varepsilon_H$)	8%	0.99	51%	0.94
w/o shrinkage	103%	0.85	99%	0.89
w/o human term	79%	0.83	103%	0.83

Table 5: Stress tests of LMR's design choices against the alternatives they replace (estimator simulation, platform-scale traffic; removal: agent-only lift removed).

Stress test (metric)	Alternative	LMR	Alt.
50% items attacked (benign weight)	current-period prior	0.90	0.82
50% attacked, 6 windows (prior $\mu_{\mathcal{G}}$)	rolling reference	0.07	0.44
Attack, then cosmetic edit (removal)	per-revision weight	79%	9%
Four small rewrites (removal)	per-revision weight	84%	20%
Ten honest revisions (weight)	plain chaining	0.85	0.69
Equal holdout size (removal)	shared 9.5% holdout	85%	68%

Removing the channel-specific allocation, shrinkage or the human term lowers removal or benign weights (Tab. 4). Without shrinkage, noisy weights remove the full lift on average but lower benign weights. Removing the human term raises removal to 79% while lowering benign weight to 0.83, and over-corrects at higher traffic (103%): the extra removal includes response that humans share. The cap at one binds on 31% of benign revisions, whose estimated gaps are negative.

6.6 Robustness and Cost (RQ5)

Across $s = 1$ to 20%, LMR leaves excess 0.7% to 4.1% against pooled 1.6% to 23.5% (Fig. 8, §C.4); at 1% both are comparable to the seed error. A 5% human holdout changes human NDCG by -0.04% and the agent holdout changes delegating-user NDCG by -0.03% (§C.4).

In the estimator simulation, each LMR design choice beats the alternative it replaces (Tab. 5). When half the items are attacked, a prior fitted to current revisions absorbs the attack gap and a rolling reference drifts toward it. Adaptive sellers who follow a rewrite with a cosmetic edit or split it into small rewrites escape per-revision weights, but chaining keeps removal at 79% and 84%; re-anchoring keeps honest sellers who revise ten times at weight 0.85 instead of 0.69. Mislabeled channels dilute the estimate: removal falls to 58% if a quarter of agent traffic goes undetected and to 56% if 5% of human traffic is labeled as agent traffic. Sparse revisions (10^3 to 10^4 impressions) keep only 40% removal (Appendix D).

Operating regime and cost. Removal rises with both the agent share and the traffic per revision (Fig. 6a, Appendix C): at $s = 1\%$, LMR removes 21% of the lift at 10^4 impressions and 66% at 10^5, whereas at $s = 20\%$ it removes 61% already at 10^4, because more agent impressions shrink the dominant agent term of Eq. 9. Benign weights stay between 0.90 and 1.00 across the grid because LMR skips low-traffic revisions (§6.4). One estimation round takes 0.1 ms for 10^3 closing comparisons and 137.1 ms for 10^6 on one CPU core (Fig. 6b); retraining dominates, and LMR leaves it unchanged.

7 Conclusion

Purchase-mediated proxy poisoning is a new threat in which sellers manipulate shopping agents through product descriptions, and genuine agent purchases then influence human-facing recommendations. We introduce GPA, which selects descriptions by a conservative surrogate gap, and LMR, which estimates the actual agent–human gap from randomized listing versions and reweights agent purchases. Our analysis characterizes how unequal responses affect shared scores, and experiments show that genuine agent purchases create excess human exposure: GPA's published rewrites raise agent selection by 70% (8.8 points) while human choices do not change measurably, and simulated pooled retraining turns this gap into 9.9%–16.2% excess human exposure across four recommenders at a 10% agent share. With platform-scale holdout traffic, LMR removes 80% of this excess without discarding useful agent purchases, leaving less than every training-side baseline that keeps them while human and delegating-user NDCG stay within 0.2%; at lower traffic it skips most revisions, and it needs to know which requests are delegated (limits in Appendix D).

Four directions remain open: surrogates, live deployment, interface protection, and undetected agents. Human-response surrogates should be evaluated across populations, product categories, and evolving shopping agents to establish their reliability. Live marketplace studies should assess the defense under changing rankings, recursive interaction histories, and repeated-user behavior. Combining purchase reweighting with interface-level protections could improve human-facing recommendations and delegated shopping outcomes. Agents that use ordinary shopping interfaces would require reliable identification of agent traffic and control over the listing versions they see.

Ethics Statement

The user studies were reviewed and approved by the authors' institutional ethics review board. All participants were adults recruited on Prolific, gave informed consent before starting, could withdraw at any time without penalty, and were paid $13 per hour plus performance bonuses. The study was described as a study of online shopping, and the debriefing disclosed that some listings had been rewritten. We collected no directly identifying information, and the released pilot records contain only anonymized choices. Because the attack is dual-use, we describe it at the threat-model level, withhold generation prompts and rewrite texts, and published no rewrite on a live marketplace; all listings were shown inside our study harness and simulated marketplaces. LMR is the countermeasure we propose for platforms, and it does not require identifying individual sellers or users.

Declaration of Generative AI

The author(s) used ChatGPT for English proofreading, reviewed all edits, and take full responsibility for the content.

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A Proofs

This appendix proves the two propositions of §4 and §5 and derives the local transfer shift of Eq. 4; notation follows Tab. 2.

A.1 Proof of Proposition 1

Proposition 1 (Finite-pool surrogate certificate). *Given the frozen pool, contexts and surrogates, with probability at least* $1-\alpha$ *under the sampling assumptions of Eq. 6, a returned candidate has* $d_a \geq \eta_A$ *and* $d_a - d_h \geq \gamma$ *for every agent and surrogate, and its gap* G *is within* $\max_{x\in\mathcal{E}}[\overline{G}(x) - \underline{G}(x)]$ *of the best eligible candidate's.*

Proof. Endpoint monotonicity of the clipped logit and nonnegative context weights give $L_m \leq d_m \leq U_m$ for every agent and surrogate, hence $\underline{G} \leq G \leq \overline{G}$ by Eq. 7. The returned candidate lies in $\mathcal{E}$, so $d_a \geq L_a \geq \eta_A$ for every agent, and $d_a - d_h \geq L_a - U_h \geq \underline{G} \geq \gamma$ for every agent–surrogate pair. Selection by Eq. 8 gives $\underline{G}(x_{j,1}) \geq \underline{G}(x^*)$, so $G(x^*) - G(x_{j,1}) \leq \overline{G}(x^*) - \underline{G}(x_{j,1}) \leq \overline{G}(x^*) - \underline{G}(x^*)$, which is at most $\max_{x\in\mathcal{E}}[\overline{G}(x) - \underline{G}(x)]$. By Hoeffding's inequality each of the M probability cells misses its interval in Eq. 6 with probability at most α/M, so the union bound makes all intervals valid simultaneously with probability at least $1-\alpha$, and every statement above holds on that event. The certificate concerns neither all fact-preserving strings nor candidates excluded by the gates. □

A.2 Proof of Proposition 2

Proposition 2 (Lift matching on fixed logged contexts). *Under (H1)–(H2), for every context* ξ *in which an agent is shown revised items and every loss* ℓ*,*

$$\mathbb{E}_{P_A}\big[w(Y)\,\ell(\xi,Y) \mid \xi\big] = C(\xi)\,\mathbb{E}_Q\big[\ell(\xi,Y) \mid \xi\big], \tag{13}$$

where $C(\xi) = 1 - \sum_j (1-w_j)\,P_A(j \mid \xi)$ *over the slate's revised items.*

If moreover (H3) one ranker parameter attains both channels' human-matched probabilities on every context of the training distribution, it minimizes both fixed-context expected losses.

Proof. Fix a context ξ and write $a_y = w(y)$ and $Z = \sum_{y'} \exp v_A(y';\xi)$ for the agents' normalizer. By (H1) and Eq. 1, $a_y P_A(y \mid \xi) = \exp(v_A(y;\xi) + \log a_y)/Z$. By (H2), $\log a_j = -g_j$ changes each revised utility from $v + \delta_{Aj}$ to $v + \delta_{Hj}$, the human-matched utility, and leaves every other utility unchanged. Normalizing gives $a_y P_A(y \mid \xi) = C(\xi)\,Q(y \mid \xi)$ with $C(\xi) = \sum_y a_y P_A(y \mid \xi) = 1 - \sum_j (1-w_j) P_A(j \mid \xi)$, and summing against $\ell(\xi,\cdot)$ yields Eq. 13; $C(\xi) \in [\min_j w_j, 1]$ when every gap is non-negative. On the fixed contexts, human outcomes and agent outcomes on slates without revised items keep weight one and are unchanged, so averaging gives the human-matched conditional-outcome loss of Eq. 3, with each agent context weighted by $C(\xi) > 0$. Under (H3), strict propriety of log-loss [10] makes the parameter that attains the human-matched probabilities minimize every conditional expected context term, including contexts without revised items, hence both positively weighted fixed-context expected losses; the statement concerns population risks, not the minimizer of a finite sample. □

A.3 Derivation of the Transfer Shift

Hold all ranker parameters except the shared item term b_j fixed, together with both channels' logged context laws. At an unregularized pooled coordinate optimum of Eq. 3, the derivative in b_j vanishes, so expected purchases of j equal predicted purchases: $(1-s)\mathbb{E}_H[\pi_H(\xi)] + s\,\mathbb{E}_A[\pi_A(\xi)] = \mathbb{E}_{\text{mix}}[p_\theta(\xi)]$. Under Eq. 1, a utility shift δ_{cj} changes $\pi_c(\xi)$ by $\pi_c(1-\pi_c)\,\delta_{cj}$ to first order, so the left side changes by $(1-s)\bar{\omega}_H\delta_{Hj} + s\bar{\omega}_A\delta_{Aj}$. A shift Δb_j changes $p_\theta(\xi)$ by $p_\theta(1-p_\theta)\Delta b_j$, so the right side changes by $J_j\Delta b_j$. Equating the two changes with $J_j > 0$ gives the first line of Eq. 4; repeating the calculation with δ_{Aj} replaced by δ_{Hj} gives Δb_j^{HM}, and subtracting gives the second line. Non-purchases already enter the score equation through p_θ and do not cancel the agent term. When the ranker is calibrated to both channels' baseline probabilities, $J_j = (1-s)\bar{\omega}_H + s\bar{\omega}_A$, and equal average curvatures reduce the excess to $s\,g_j$.

A.4 Estimated and Clipped Weights

With estimated and clipped weights, the normalized agent choice distribution of Proposition 2 keeps a residual cumulative utility gap

$$\Gamma_j - \min\{\max\{\Sigma_j, 0\}, -\log w_{\min}\},$$

where Γ_j is the true cumulative gap and Σ_j its estimate in Eq. 12. With known hyperparameters, a single revision's estimation error under Eq. 11 has prior-averaged mean square $\tau_G^2\sigma_j^2/(\tau_G^2 + \sigma_j^2)$ [30]; §6.5 measures the residual empirically.

B Implementation Details

This appendix completes the setup of §6.1.

B.1 Harness and GPA Instantiation

Items are the most-reviewed items with complete attributes in each subcategory, a rule fixed before observing outcomes. A preregistered goal score from fixed attributes defines the goal-optimal item; goals whose two best scores fall within a preset margin are discarded. The audit's one-sided 95% upper bound on the violation rate is 9.1%. The pilot's 200 Prolific participants made 2,400 choices between original and alternative descriptions in subcategories disjoint from the evaluation, released as public records. For each item a generator (Gemma 3 27B, temperature 0.7) proposes $C_{\max} = 8$ candidates; a rule-based attribute matcher marks each pass, fail or unsure, and the same model judges unsure spans. Of the candidates, 6.1 on average pass the validator and lie within the surrogates' training support; the two scoring agents answer 2,048 times per text and context on $T = 8$ contexts, with $\alpha = \zeta = \eta_A = 0.05$ and

$\gamma = 0.1$, at temperature 1 in fresh sessions. Targets in the retraining worlds are drawn among items with a published revision; items where GPA keeps the original stay unrevised, as a seller would deploy only accepted rewrites. GPA and max-share selection pick the same text for 62% of published items. Real rewrites move agents far less than the banks of §C.1, which is why real listings need $n = 2{,}048$. Generation prompts and rewrite texts are withheld, and no rewrite was published on a live marketplace.

B.2 User Study

Eligible participants are US residents aged 18 or older who shop online at least monthly and use a desktop browser. Tasks come from both datasets and use items with a published GPA revision (GPA runs before both studies), allocated equally across the 5 conditions, and no participant sees two versions of the same listing. Participants receive $3.25 for an estimated 15 minutes ($13 per hour) and $0.10 for each choice of the goal-optimal item. We exclude and replace participants who fail either of two attention checks, finish in less than a third of the median time, or report, without penalty, using an AI assistant; human intervals are clustered by participant.

B.3 Choice Model and Traffic

For channel c, the utility of item j in version z at slate position m for user u is $v_c = \beta_{cj}(z) + \mathbf{p}_u^\top \mathbf{q}_j + \kappa_c \log(1/\log_2(m+2))$, with user and item factors fitted to review ratings, item–version effects $\beta_{cj}(z)$ and channel position sensitivity κ_c ($\kappa_A = 0$). On held-out 10-item agent tasks, a transfer test from the 8-item harness slates [44], predicted selection shares differ from observed ones by 1.2 points on average. The leave utility is set so that 50% of requests end in a purchase (5% per impression). A revised item receives a median of 250 impressions per window in the simulated marketplaces; each simulated impression supplies 200 platform impressions to the default holdout, while ranker training uses the simulated events. The full-scale world multiplies users and requests by 200, so the holdout counts and the training events coincide; it runs Pooled, Holdout only and LMR with LightGCN (seed SE about 2.4 points with three seeds; LMR changes delegating-user NDCG by -0.18%). The ML-1M subset has 1,500 users and 97,798 ratings, with a parametric agent model calibrated to RQ1.

B.4 Recommenders, Policies and Metrics

In SASRec and UniSRec the weight scales the loss at the purchased position, and the purchase stays unweighted in later input histories; during a holdout UniSRec encodes the current version. All policies share architecture, epochs, event budget and tuning trials; open-weight agents (Qwen3-235B-A22B with 4-bit AWQ weights) and all rankers run on four NVIDIA A100 80GB GPUs; scoring calls return one item index and reuse cached prompt prefixes. LMR uses $\varepsilon_A = 50\%$, $\varepsilon_H = 5\%$, $w_{\min} = 0.1$, $\tau_{\min} = 0.1$, $\sigma_{\max} = 1$ and $M_a = 4$. The global weight is chosen from $\{0, 0.25, 0.5, 1\}$ each round (validation loss selects one in most rounds); clipping keeps at most 3 agent purchases per item, user and round; LoRec's user-wise weights are adapted to the slate loss; canonicalized text rewrites seller text into a fixed template, and the facts-only view shows agents the structured attributes. The evaluation model predicts held-out choices with top-1 accuracy 31% (chance 12.5%) and calibration error 1.8 points; paired NDCG differences have a seed SE of about 0.03%. Fake-user injection buys each attacked item as often as the attack's additional agent purchases, with popularity-drawn fillers; the detector uses standard shilling profile features and is trained on injected profiles from separate runs.

C Additional Experiments

C.1 Selector Verification

This study extends RQ1 (§6.2) by isolating GPA's selection rule from generation and model behavior. Three known Bernoulli banks hold the original and four candidates, two agents, two contexts and two deterministic human surrogates; in the separation bank, from probability 0.2, a *shared* candidate raises agents and surrogates to 0.6, a *specific* one raises agents to 0.5 only, a *brittle* one raises a single agent, and a *negative* one lowers both. Each bank is scored at five budgets $n \in \{16, \dots, 4096\}$ with 200 replicates, using the settings of Appendix B.

At $n = 4096$, GPA selects the agent-specific candidate in all replicates and attains the known surrogate gap $\log 4$, while max-share selection picks the shared improvement with zero gap (Fig. 7, §6.2); GPA publishes nothing at $n \le 64$ and 89% of the time at $n = 256$. In the heterogeneous bank (unequal context weights and agents), full GPA attains gap 1.233 versus 0.191 for single-agent GPA. In the misspecified-human bank, where only the specific candidate's human response changes to 0.8, GPA still selects it: the surrogate gap stays $\log 4$ while the true gap is $-\log 4$.

C.2 Full Defense Comparison

Tab. 6 lists the values behind Fig. 4 (RQ3, §6.4), with human NDCG and LMR at the simulated marketplace's own and 20 times its traffic.

Table 6: Defense comparison, AMZ-E/LightGCN, round 3: excess exposure, NDCG change from no-attack pooled training (paired SE about 0.03%); indented: less holdout traffic.

	Policy	Excess	Human	Deleg.
Training	Pooled	13.9%	-0.21%	-0.18%
	Drop	0.0%	-0.55%	-1.20%
	Global weight	13.4%	-0.20%	-0.19%
	Channel heads	10.7%	-0.16%	-0.20%
	Clipping	9.9%	-0.19%	-0.24%
	LoRec	13.5%	-0.23%	-0.21%
View	Canonicalized text	6.4%	-0.10%	-0.27%
	Facts-only view	0.0%	-0.09%	-0.62%
	LMR	2.7%	-0.11%	-0.19%
	20× traffic	7.1%	-0.15%	-0.19%
	1× traffic	9.6%	-0.21%	-0.20%
	Holdout only	11.6%	-0.21%	-0.20%
	Oracle	0.5%	-0.02%	-0.02%

C.3 Estimator Details

These results extend RQ4 (§6.5). Under repeated-user assignment, a user-cluster-robust variance replaces σ_j^2 of Eq. 10. Log odds do not collapse over contexts, so the pooled $\hat{\delta}_c$ targets the utility shift only for small, similar purchase probabilities; otherwise it is estimated within predeclared strata. At the default traffic, the median-revision standard error of Eq. 9, 0.16, matches the plug-in value 0.16. Spreading each revision's impressions over 5 strata with baseline log-odds spread 1.0, the pooled estimate recovers 95% of the attack gap at a

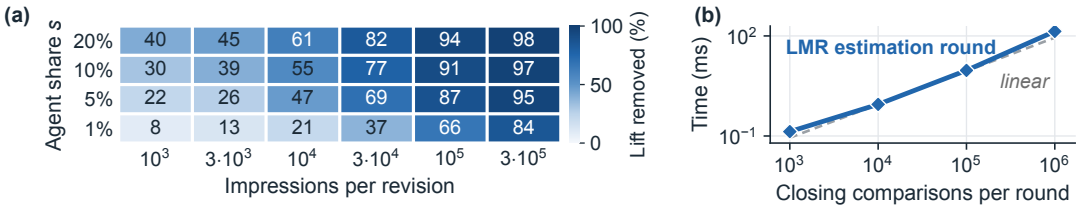


Figure 6: (a) Agent-only lift removed (%) by LMR's per-channel holdout over agent share and impressions per revision (mean of 200 categories). (b) Time of one estimation round (gap, shrinkage, weights) against the number of closing comparisons; dashed: linear scaling.

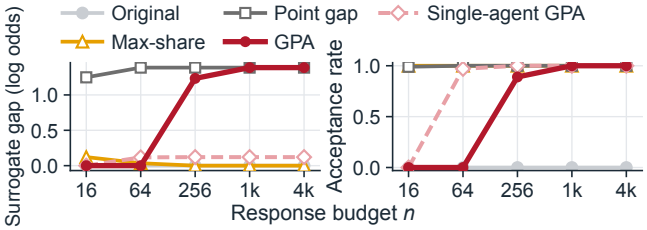


Figure 7: Selector verification, separation bank: known surrogate gap and acceptance across response budgets, over 200 replicates per setting. Max-share picks the highest target-choice share; point gap, the largest estimated gap without bounds; single-agent GPA scores one agent.

5% purchase rate and removes 82%, against 89% for stratified estimates; at 20% the removals are 82% and 95%. At 250 impressions per revision, correcting every revision removes 97% but lowers every revised item's weight, benign ones included, to about 0.60. The floor binds on 0.0% of weights; removing the cap raises the mean benign weight from 0.91 to 0.94.

C.4 Robustness Details

These results extend RQ5 (§6.6); Fig. 8 plots the agent-share sweep.

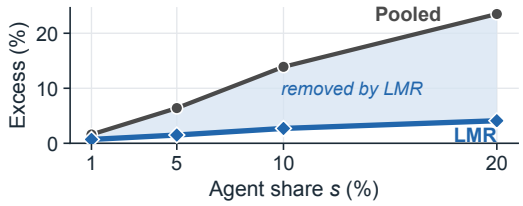


Figure 8: Excess human exposure of attacked items against the agent share s (AMZ-E/LightGCN, round 3); the shaded band is the exposure LMR removes.

In the estimator simulation, at the simulated marketplace's own traffic and at 20 and 200 times it, LMR removes 19%, 43% and 85% of the lift. Across $s = 1$, 5, 10 and 20%, lift removal is 49%, 80%, 85% and 88%. At attack shares of 5, 20 and 50%, the fixed reference gives benign weights 0.91, 0.90 and 0.90, versus 0.90, 0.86 and 0.82 with a current-period prior. Human holdout shares of 1%, 5% and 10% change human NDCG by -0.01%, -0.04% and -0.08%, and the agent holdout changes delegating-user NDCG by -0.03%.

D Discussion

Scope of the certificate. GPA's certificate (Proposition 1) covers the frozen pool and the declared surrogates, not all fact-preserving texts or the platform's shoppers. Surrogate specificity does not imply human specificity: in §C.1, a misspecified human surrogate reverses the selected gap's sign even with precise agent sampling. Only held-out calibration on independent data licenses a claim about a real human gap.

Evaluation scope. The retraining worlds are semi-synthetic: human and agent choices from the harness and the user study are extended to new slates by fitted logit models that assume agents ignore slate position, and the transfer and defense estimates hold under them. The default holdout draws 200× the simulated traffic independently from the fitted models, one week of a 300-item category served 1,600,000 requests (50,000 impressions per item); duplicating each impression 200 times would make the plug-in standard error 2.8 times smaller than the actual error. Harness and study tasks require a purchase, so RQ1 measures the gap conditional on buying; the retraining worlds add a calibrated leave utility, and LMR's holdout counts purchases per impression. The baseline comparison and full-scale check use AMZ-E/LightGCN at $s = 10\%$, and max-share comparisons one pool per item and 3 backbones.

Estimator assumptions. The estimator study draws impressions independently, and its repeated-revision cases regenerate each comparison's baseline; repeated-user assignment needs the cluster-robust variance of §C.3. The reference period must be free of attacks, which a platform can only approximate, and revisions with fewer than about 10^4 impressions per window are shrunk strongly toward the category's typical gap; below about 10^3, LMR skips most of them. At the default traffic, LMR removes 40% of the agent-only lift for revisions with 10^3 to 10^4 impressions and 92% above 10^5.

Deployment. LMR needs to know which requests are delegated: undetected agents' purchases keep weight 1, and mislabeled traffic dilutes the estimated gap. Removal falls from 85% to 74%, 58% and 35% when 10, 25 and 50% of agents go undetected, and to 77% and 56% when 1 and 5% of humans are labeled agents. Sellers can time revisions to low-traffic weeks or revise within one window, an edit paired with a genuine attribute change bypasses the holdout, and a new listing has no predecessor to randomize against. LMR also makes a normative choice: human-facing rankings absorb only the effect humans share, even when agents' extra response is legitimate.